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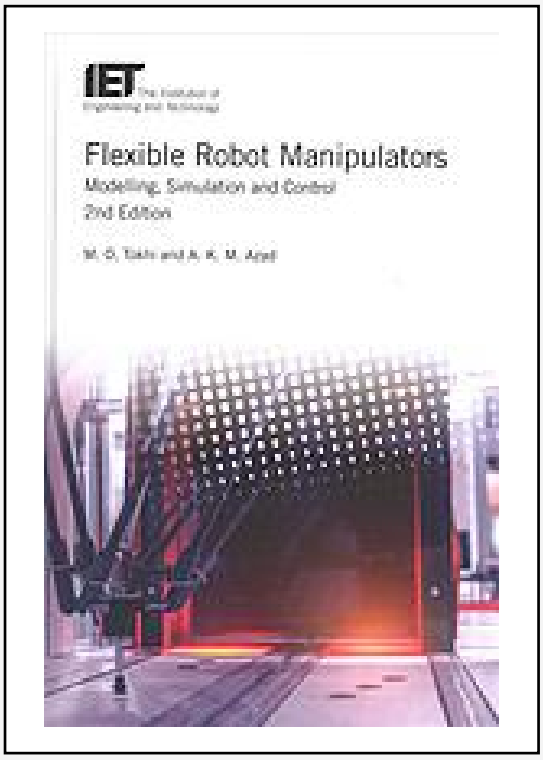
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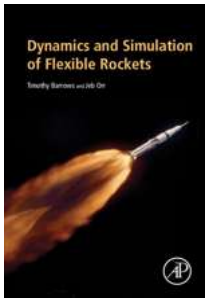
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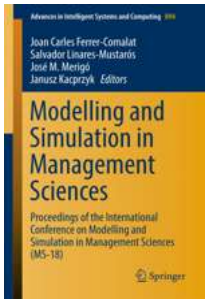

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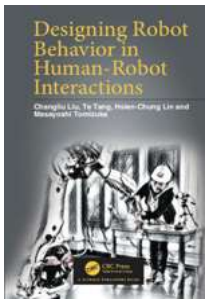

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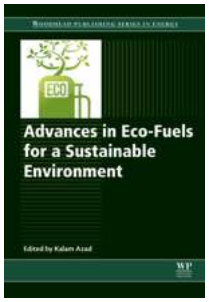

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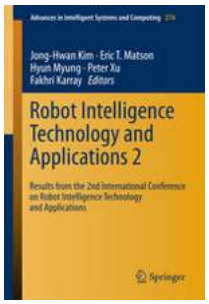

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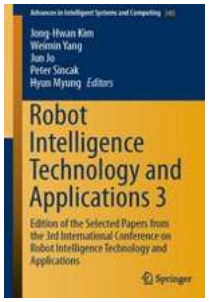

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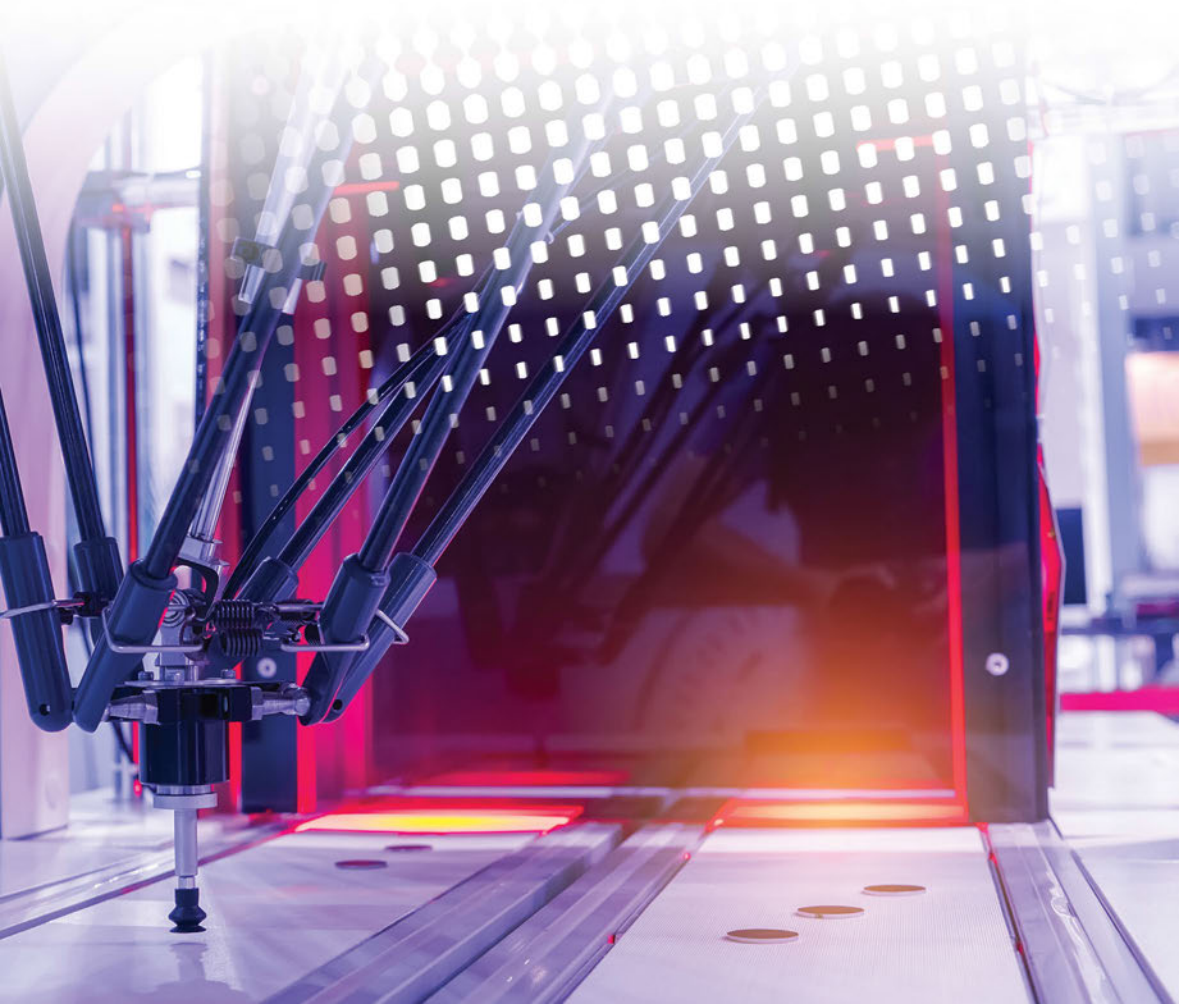
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## 2nd Edition

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# Foreword

The ever-increasing utilisation of robotic manipulators in various applications has been motivated by the requirements and demands of industrial automation. Among the rigid and flexible manipulator types, attention is focused more on flexible manipulators. This is due to various advantages such manipulators offer when compared to their rigid counterparts. Exploitation of the potential benefits and capabilities of rigid and flexible manipulators introduces a further emerging line of research in which hybrid rigid-flexible manipulator structures are considered.

Flexural dynamics (vibration) in flexible manipulators has been the main research challenge in the modelling and control of such systems. Accordingly, research activities in flexible manipulators have looked into the development of methodologies to cope with the flexural motion dynamics of such systems.

A considerable amount of research on the development of dynamic models of flexible manipulators has been carried out. These have led to descriptions in the form of either partial differential equations, or finite-dimensional ordinary differential equations. From a control perspective, an input/output characterisation of the system is desired, which can be obtained through suitable on-line estimation and adaptation mechanisms. Given the dynamic nature of flexible manipulator systems, the practical realisation of such methodologies present new challenges.

Numerical techniques using finite difference and finite element methods have been researched for dynamic characterisation of flexible manipulators. Accordingly, simulation algorithms characterising the dynamic behaviour of flexible manipulators have been developed which provide flexible means of analysis, test and verification of control techniques. With the widely available use of digital computing technology, such platforms are first-step favoured option in a wide range of applications.

Control structures adopted for flexible manipulators can broadly be grouped into open-loop and closed-loop types. Although, the mathematical theory of open-loop control is well established, only a limited number of successful applications in the control of distributed parameter flexible manipulator systems have been reported. A further research dimension, with this class of control structures, is on-line adaptation of the input shaping mechanism with the changing behaviour of the system and the environment. With closed-loop control techniques, a common trend that has been adopted by researchers is partitioning of the dynamics of the system into the slow (rigid-body) and fast (flexural motion) dynamics and accordingly devising separate control loops. An important consideration with this has been to adequately cope with the non-minimum phase behaviour exhibited by the system characterisation, which with optimal feedback control techniques leads to unstable control. Although

this problem can be avoided with some traditional techniques, emerging intelligent control methodologies incorporating soft computing paradigms offer a great deal of potential in solving such problems.

This book reports recent advances and new developments in analysis (modelling and simulation) and control of flexible robot manipulators, in the light of the issues mentioned above. The contents of the book are presented in 15 chapters, and these may be categorised into five parts. Part 1 comprises Chapters 1 and 2, and provides a general overview of flexible manipulators from the perspective of modelling, simulation, control and applications a design formulation of flexible manipulators. Part 2 is on modelling and simulation techniques. It includes Chapters 3–7, and presents classical modelling techniques based on the Lagrange equation formulation, parametric approaches based on linear input/output models using system identification techniques, neuro-modelling approaches, and numerical modelling/simulation techniques for dynamic characterisation of flexible manipulators using finite difference and finite element techniques. Part 3 is on classical and advanced control techniques. Chapters 8–10 are grouped into this section, and includes a range of open-loop and closed-loop control techniques based on classical and modern control methods including neuro and iterative control. Part 4 includes Chapters 11–14, and presents a range of soft-computing control techniques based on fuzzy logic, neural networks, and evolutionary and bio-inspired optimisation paradigms. Part 5 includes Chapter 15, and presents a software environment for analysis, design, simulation and control of flexible manipulators. A range of approaches presented in earlier chapters are brought together in this chapter into an interactive software environment for use in both in research and education.

The book is intended for teaching in graduate courses on robotics, mechatronics, control, electrical and mechanical engineering. It can also serve as a source of reference for research in areas of modelling, simulation and control of dynamic flexible structures in general and, specifically, of flexible robotic manipulators.

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# Abbreviations

| | |
|---|---|
| A/D | Analogue/digital |
| ADAM | Aerospace dual-arm flexible manipulator |
| ARMAX | Autoregressive moving average with exogeneous inputs |
| ARX | Autoregressive with exogenous inputs |
| AVQ | Adaptive vector quantisation |
| CACE | Computer aided control engineering |
| CCD | Charged coupled device |
| CS | Command shaper |
| D/A | Digital to analogue |
| DOF | Degrees of freedom |
| EAP | Electroactive polymer |
| EI | Extra insensitive |
| EMI | Electromagnetic interference |
| ERLS | Equivalent rigid link system |
| FD | Finite difference |
| FE | Finite element |
| FLC | Fuzzy logic control |
| FRF | Frequency response function |
| GA | Genetic algorithm |
| GOCF | Generalised observability canonical form |
| IAE | Integral of absolute error |
| ILC | Iterative learning control |
| IMSC | Independent modal space control |
| ISE | Integral of square error |
| ITAE | Integral of time weighted absolute error |
| JBC | Joint-based collocated |
| LMS | Least mean squares |
| LQR | Linear quadratic regulator |
| LRMS | Long-reach manipulator system |